

BRAMBLELOOP STUDIO

AUTONOMOUS CROCHET COMMERCE OPERATING SYSTEM

One Master Build Specification for Claude - v1.1

MISSION: Build a premium crochet publishing company with a credible path to CA\$100,000+ annual revenue. Claude builds the persistent cloud company OS; the OS runs continuously when Claude and the owner computer are off. The owner approves only genuine owner-only actions.

1. Business Thesis and CA\$100K Target

Brambleloop Studio is a premium, Etsy-first digital crochet publishing company powered by a persistent cloud operating system. Specialized agents research demand, create original concepts, engineer machine-readable patterns, validate them, publish premium products, market them, support buyers and learn from every outcome.

North star: qualified demand x click-through x conversion x average order value x repeat purchases - platform fees - ads - AI/hosting/testing/support costs = sustainable net profit.

CA\$100,000/year is about CA\$8,333/month. At CA\$17 blended AOV that is roughly 5,883 orders/year, 490/month or 16/day. This is a target, not a guarantee. Initial price test bands: singles CA\$5.99-14.99, flagships CA\$11.99-16.99, mini collections CA\$17.99-29.99 and major collections CA\$29.99-59.99. Pricing Intelligence replaces these assumptions with evidence.

2. Pattern Engineering - Non-Negotiable Quality

PATTERNS ARE SOFTWARE RELEASES. Never create a beauty image first and ask an LLM to guess instructions. Every product begins as a formal Crochet Intermediate Representation (CIR).

Release pipeline: Market Brief -> Original Design -> CIR -> Deterministic Compiler -> Digital Twin -> Charts -> Written Pattern -> Reverse Compiler -> Adversarial QA -> Physical Test if required -> Asset Truth -> Policy Gate -> Quality Release Certificate.

CIR stores version, construction, materials, gauge, geometry, rows/rounds, repeats, colour changes, shaping, motifs, components, assembly, grading, finishing and evidence. Any unresolved arithmetic error, invalid repeat, chart mismatch, reverse-compile mismatch, unsupported claim, material image mismatch, IP concern or policy failure blocks release.

3. Digital Twin, Reverse Compiler and Physical Testing

Critical arithmetic is deterministic code, not LLM opinion. Validate every row/round start/end count, repeats, motif dimensions, increases, decreases, joins, colour sequences and foundation compatibility. The Digital Twin executes canonical operations and renders structural output. A separate Reverse Compiler sees only the customer pattern and must reconstruct CIR.

Track Arithmetic, Instruction, Chart, Gauge, Yardage, Visual and Physical confidence separately. Risk A flat/mosaic/granny work can rely heavily on computation; Risk B bags/hats/simple amigurumi need more physical sampling; Risk C fitted garments and complex structures normally require physical testing. Tester results become calibration data for future patterns.

4. Market Radar and Competitor Digital Twins

Continuously scan blankets, mosaic, graphghans, amigurumi, stuffed animals, baby, hats, scarves, sweaters, cardigans, tops, dresses, shawls, bags, baskets, pillows, runners, placemats, coasters, wall decor, ornaments, stockings, tree skirts, flowers, pets, nursery, wedding, gifts, seasonal decor, quick makes, low-sew/no-sew and emerging niches.

Maintain longitudinal profiles for serious competitors: catalogue mix, launches, visible bestseller/review signals, pricing, promotions, bundles, hero photography, recurring model use, video, free-vs-paid content, launch timing, websites, YouTube/Pinterest, praise and complaints. Agents: Discovery, Digital Twin Builder, Change Detector, Review Miner, White-Space, Category Analyst, Trend Forecaster and Seasonality Engine. Learn why winners work; never copy protected expression.

5. Seasonality, Launch Events and Winner Amplification

Model SHOPPING DATE separately from MAKING DATE. Maintain a 365-day calendar across major holidays, weddings, baby, graduation, cottage season, back-to-school, fall, winter and evergreen gifts.

Flagship timeline: T-90 market thesis; T-75 concept; T-60 CIR; T-45 QA/testing; T-30 final product/tutorial/assets; T-21 SEO/Pinterest indexing; T-14 teasers/email/social; launch; post-launch tutorial/CAL/content, optimization, retargeting and bundles.

Build reusable original validated motif/component libraries. Winner Amplification turns proven demand into matching products, bundles, sizes, colourways, tutorials, content and controlled ad expansion.

6. Brand System, Storefront and Consistent Model

The Etsy grid must look like one premium design house. Lock palette, typography, crop rules, lighting, editing, information-card style, collection naming, PDF design and visual QA.

Create one fictional adult female Brambleloop brand model for appropriate wearable/accessory imagery: glamorous brunette, striking light eyes, strong brows, refined symmetrical features, polished editorial beauty, warm approachable expression and tasteful styling. Evoke a high-glamour brunette aesthetic without duplicating a real celebrity. Maintain a character bible so face, hair, proportions and styling stay consistent. Use product-first imagery for blankets/decor when stronger.

Storefront Director owns banner, icon, About, policies, sections, announcement, thumbnails and grid coherence.

7. Listing Conversion and Asset Truth

The first image exists to win the click. Default to a clean finished-result hero, not a cluttered collage. Later frames explain value: lifestyle, detail, second result/model, what is included, difficulty/time, materials, sizes, pattern preview and collection cross-sell.

Asset classes: PHYSICAL_PRODUCT_PHOTO, DIGITAL_TWIN_RENDER, AI_LIFESTYLE_CONCEPT, INFOGRAPHIC and PATTERN_PREVIEW. Asset Truth Engine blocks motifs, textures, shapes, garment construction or scale that materially misrepresent the pattern. Thumbnail Warfare Agent evaluates hero variants at mobile/search-grid scale using real CTR, favorites and conversion.

8. Search Domination and Page Growth

Growth objective: qualified Etsy search share through MATCH -> CLICK -> CONVERT -> DELIGHT. Search Domination Agent builds query coverage for every product/category/season, uses accurate categories and attributes, all 13 relevant tag opportunities, clear buyer-friendly titles, strong descriptions and complete shop information.

Search Visibility Monitor converts Etsy recommendations into jobs. Launch Velocity coordinates legitimate early traffic so new listings gather real engagement. External Traffic builds Pinterest, Google/SEO, YouTube, email, social and creator traffic.

Hard rule: no fake reviews, fake purchases, fake favourites, sock-puppet accounts or fabricated engagement. Growth must create real buyers and real satisfaction.

9. Pricing Intelligence and Promotions

Create Market Price Scanner, Price Positioning Agent, Promotion Agent, Price Experiment Agent, Bundle Economist and Revenue Optimizer. Observe comparable price, complexity, charts/video, seller maturity, category, season, bundle size, visible promotion and our conversion.

Optimize expected contribution profit per visitor, not cheapest price or gross revenue. Maintain genuine regular prices. Do not create deceptive perpetual sales or manipulate reference pricing. Promotion logic must pass Policy Agent review. Bundle pricing should lift AOV while preserving obvious customer value.

10. AI Customer Experience Department

Customer Concierge triages every authorized message. Specialists: Download Support, Crochet Help, Materials, Pattern Troubleshooter, Customer Happiness, Incident Agent and Review Intelligence. Pattern answers use the exact canonical version; agents cannot invent stitch corrections.

Repeated questions about the same row/round create a potential defect incident. P0 impossible and P1 major defects can pause listings, rerun QA, patch CIR/PDF, increment versions and prepare affected-customer communication. Routine questions should not wake the owner. Legal/privacy/chargeback/unusual disputes escalate.

Customer service is also growth: fast responses, low case rates and legitimate strong reviews improve trust and can support Etsy visibility.

11. Marketing and Paid Media

Agents: Chief Growth, Pinterest Organic, Pinterest Paid, Etsy Ads, Organic Social, Email, SEO/Content, Video/YouTube, Creative Testing, Attribution and Creator/Partnership Research. Every strong product becomes a content ecosystem: Etsy listing, website article, Pinterest variants, tutorial/video, short clips, email, teaser/free motif where useful, collection cross-sell and crochet-along candidate.

Paid Media has hard daily, campaign and monthly caps plus max test loss, max CAC, target contribution/ROAS and minimum-data thresholds. Start small; scale only profitable evidence. Auto-pause on tracking failure, unavailable listing, CAC breach, refund anomaly or test-loss cap. CFO/Skeptic independently reviews increases; consequential spend requires owner approval.

12. Portfolio, Reviews and Product Mortality

Track impressions, clicks, CTR, favorites, conversion, orders, revenue, refunds, support incidents, reviews, ad spend, contribution margin, keyword performance and repeat/cross-sell for every SKU. Classify STAR, PROMISING, SEO_PROBLEM, CONVERSION_PROBLEM, QUALITY_PROBLEM, SEASONAL, REWORK or RETIRE.

High impressions/low clicks = hero/title problem. High clicks/low conversion = offer/trust/price. Low impressions/high conversion = distribution opportunity. High sales/high support = quality emergency. Review Intelligence mines genuine feedback for product requirements and clarity issues. Persistent losers are retired after defined intervention ladders.

13. 24/7 Cloud Architecture

Claude builds a persistent cloud application; Claude itself does not need to remain open. Core: admin dashboard, API, Postgres, object storage, durable job queue, orchestrator, specialist workers, cron/event scheduler, secrets manager, logs, monitoring and backups. Jobs include ID, agent, type, inputs, priority, status, lease, attempts, timestamps, errors, retry, cost, outputs and idempotency key.

Cadence: 15-minute infrastructure heartbeat; hourly queue/incident checks; 6-hour meaningful market/store/ad signal review; daily sales/traffic/pricing/ads/support/competitor/pipeline/cash review; weekly portfolio allocation and CFO challenge; monthly strategy recalibration and CA\$100K trajectory. Agents sleep when no valuable work exists.

14. Agent Governance, Policy and Owner Approval

Each agent gets explicit tools, token/API budget, daily cost ceiling, retry limit and financial authority. Listing cannot spend ad money. Marketing cannot alter CIR. Designer cannot publish. Support cannot silently patch canonical patterns. Pricing cannot bypass Policy. No single model can bypass release gates.

CEO/Orchestrator sets priorities; Quality Director owns release certificates; CFO/Skeptic challenges spending; Policy & Platform Compliance owns current Etsy/platform checks. Never automate KYC, CAPTCHA bypass, identity verification, legal acceptance or unsupported credential actions.

Owner receives one consolidated OWNER ACTION REQUIRED queue with exact action, reason, maximum cost, minutes required and consequence of delay.

15. Intelligence Warehouse, Finance and Failure Recovery

Persist PRODUCTS, COLLECTIONS, PATTERN_VERSIONS, CIR_NODES, ROWS/ROUNDS, MOTIFS, MATERIALS, GAUGES, VALIDATION_RUNS, PHYSICAL_TESTS, ASSETS, LISTINGS, KEYWORDS, COMPETITORS, SNAPSHOTS, SALES, REVIEWS, SUPPORT_CASES, INCIDENTS, EXPERIMENTS, CAMPAIGNS and FINANCIAL_LEDGER.

Track gross sales, discounts, refunds, platform/payment fees, ads, AI/API, hosting, software, testers/contractors, tax reserve, contribution margin, net profit and cash. Capital Allocator sends resources toward highest expected marginal contribution.

Heartbeat detects dead workers, expired integrations, tracking failure, duplicate jobs, runaway token cost, corrupted artifacts, failed uploads and support spikes. Use retries/backoff, circuit breakers, dead-letter queues, idempotency, backups, immutable release hashes and audit logs.

16. Initial Catalogue and Full Agent Organization

Begin with about 12 exceptional release candidates, not hundreds of mediocre products. Favor machine-verifiable seasonal mosaic/colourwork, home decor, runners/pillows, selected blankets, accessories and carefully chosen simple amigurumi while validation matures. Reallocate aggressively toward observed winners.

Executive: CEO/Orchestrator, CFO/Skeptic, Policy, Quality Director. Intelligence: competitor, category, trend, seasonality, keyword/search. Creative: Creative Director, Collection Architect and category specialists. Engineering: Crochet Engineer, CIR Compiler, Validator, Digital Twin, Reverse Compiler, Adversarial Reviewer, Gauge/Yardage and Test Coordinator. Brand/Publishing: Storefront Director, Brand Model Consistency, PDF/Charts, Visual Director, Asset Truth, Listing, Thumbnail Warfare, Etsy SEO. Commercial/Growth: Pricing, Promotion, Bundles, Revenue, Store Operator, Portfolio, Search Domination, Launch Velocity, Growth, Pinterest, Etsy Ads, Social, Email, SEO, Video and Attribution. Customer Experience: Concierge, Download, Crochet Help, Materials, Troubleshooter, Happiness, Incident and Review Intelligence. Operations: Scheduler, Queue, Integration Health, Cost, Backup/Recovery and Owner Action Queue.

17. Final Release Chain and Living Master Plan

Market Radar -> Opportunity Score -> Creative Brief -> Originality/IP Gate -> CIR -> Compiler -> Simulator -> Reverse Compiler -> Adversarial QA -> Physical Test if required -> PDF/Charts -> Asset Truth -> Pricing -> Listing/SEO -> Policy Gate -> Quality Release Certificate -> Store Operator -> Marketing -> Customer Experience -> Review/Incident Learning -> Portfolio Reallocation.

This document is canonical v1.1. Whenever material market evidence, policy changes, QA improvements, profitable channels, pricing lessons, customer-service insights or failure lessons improve the architecture, increment the version, regenerate the PDF, preserve a changelog and provide the newest master copy.

v1.2 - FINAL PRE-CLAUDE ADVERSARIAL AUDIT

This addendum is part of the canonical master plan. It closes the remaining architecture gaps found after reviewing the current Etsy crochet search landscape and Etsy's current search/policy guidance.

25. Search Reality Corrections

Do not treat Star Seller or Etsy Ads as organic-ranking hacks. Current Etsy guidance says Star Seller is a trust signal but does not directly order organic results; Etsy Ads do not improve organic placement outside designated ad inventory.

Search agents must focus on query matching, listing relevance, first-photo click-through, engagement, conversion, listing/shop quality and customer-service performance.

Maintain a query coverage map and ingest Etsy Search Visibility recommendations whenever accessible. Use all relevant categories/attributes and all 13 tag opportunities accurately; avoid keyword stuffing.

Current crochet search results show broad winning demand across premium pattern+video garments, mosaic/blankets, baby items, no-sew/low-sew amigurumi, bags, quick market-prep projects and bundles. The portfolio allocator must remain category-agnostic and evidence-driven.

26. Shadow Mode and Safe Autonomy

Claude must not connect a newly built autonomous system directly to live customers, live listings and live ad spend on day one.

Phase 1 SHADOW MODE: agents research, design, validate, price, draft listings, simulate customer support and propose campaigns without publishing, messaging or spending.

Phase 2 STAGING: end-to-end rehearsals using test data and non-production artifacts.

Phase 3 LIMITED PRODUCTION: enable one capability at a time after acceptance tests pass.

Authority matrix: GREEN = autonomous reversible/no-spend actions; YELLOW = bounded customer/spend actions inside explicit limits; RED = owner-only KYC, legal acceptance, banking, consequential spend and irreversible/high-risk actions.

Every production action must record agent, model/prompt version, policy version, inputs, outputs and timestamp.

27. Model Gateway and Reproducibility

Do not hard-wire the company to one LLM provider. Put all model calls behind a Model Gateway.

Route tasks by capability, reliability and cost; support provider outage failover; pin critical model/prompt versions; log provenance and cost.

Critical pattern validation remains deterministic even if every LLM provider disagrees.

Maintain evaluation fixtures for research extraction, listing copy, support answers and pattern reconstruction so model upgrades can be benchmarked before production rollout.

28. Security, Privacy and Disaster Recovery

Use least-privilege credentials, server-side secret storage, encryption in transit, role-based access and immutable audit logs.

Separate production, staging and test environments. Never use real customer data in model tests unless required, permitted and minimized.

Back up database, CIR artifacts, release PDFs, listing assets, financial ledger and configuration. Periodically perform a restoration drill; a backup that has never been restored is unproven.

Create incident playbooks for leaked credentials, compromised integrations, accidental customer-data exposure, corrupted pattern artifacts and unauthorized ad spend.

29. Originality, Trademark and Brand Clearance

Add an IP/Originality Agent before creative release. It checks proposed names, phrases, characters and designs for obvious collision risk and records sources reviewed.

Do not assume the working name Brambleloop Studio is legally clear merely because it sounds good. Before commercial launch, perform marketplace/domain/social-handle screening and appropriate trademark clearance; escalate formal legal clearance to the owner/professional if needed.

Never create products based on protected characters, celebrity likenesses, logos or franchise identifiers merely because competitors do so.

30. Owned Audience and Channel Resilience

Etsy is the launch channel, not the entire company. Build an owned website, SEO library and consented email audience.

Every free article/tutorial must have a commercial job: rank for a useful query, demonstrate expertise, capture appropriate consent, support a paid product or reduce customer friction.

Maintain channel-attribution data so Pinterest, Google, YouTube, email, Etsy organic and paid traffic can be compared on contribution profit.

31. Localization and Accessibility

Design the product model so customer-facing patterns can later support additional languages without forking the canonical CIR. Translation is downstream of CIR and requires terminology review before sale.

PDFs, charts and web assets should prioritize legibility, contrast, clear legends, print-friendly versions and colour-independent cues where practical.

Accessibility is both customer experience and support-cost reduction.

32. Pre-Launch Acceptance Tests

Before the owner is asked to connect the live Etsy store, Claude must demonstrate: durable queue recovery after worker restart; duplicate-job protection; database backup and restore; model-provider failover; spend-limit enforcement; owner-action escalation; CIR compiler tests; reverse-compiler mismatch detection; PDF generation; asset-truth blocking; policy veto; support incident creation; and financial ledger reconciliation.

Run a full simulated product from market discovery through release certificate without manual intervention.

Run a deliberately broken pattern and prove the system refuses publication.

Run a deliberately misleading listing image and prove Asset Truth blocks it.

Run an ad-budget breach simulation and prove the campaign is prevented/paused.

Run a repeated customer defect simulation and prove it creates a correlated incident.

33. Initial Launch Portfolio Test

Do not commit the first 12 SKUs before Market Radar runs. Start with a candidate pool of at least 30 concepts across several categories, score them, then choose approximately 8-12 release candidates.

Favor Class A/B products initially while the validation system accumulates physical calibration data.

Require at least one flagship seasonal product, several faster lower-price products, at least one bundle-ready family and at least one evergreen search product so early data is not dependent on a single demand pattern.

Do not spend heavily on ads until listing quality and conversion have some evidence; bounded discovery tests are allowed.

34. Cost-to-Create and Throughput Metrics

Track agent/API minutes and dollar cost per concept, validated pattern, PDF, listing, video asset, support case and acquired customer.

A six-figure store can still be a bad business if autonomous production burns excessive API and ad spend.

CEO/CFO must optimize profitable throughput: validated products per dollar and contribution profit per operating dollar.

35. What Must Happen Before Sending Claude Live

Nothing else needs to be invented at the strategy level before Claude starts building. The next unknowns are implementation discoveries, not reasons to keep planning indefinitely.

Send Claude this complete v1.2 master plan and instruct it to build in phases without asking routine questions.

Owner-only setup can be deferred until the system reaches the integration that genuinely needs it: Etsy account/KYC, banking/payment details, any paid infrastructure approval, and final brand/trademark decision.

Claude should first build the core cloud architecture, database, queues, CIR/compiler/validator, admin dashboard, Market Radar, shadow-mode workflow and release gates. Live Etsy publishing and paid ads come later.

Require Claude to maintain a BUILD_STATE record and checkpoint after every coherent milestone so another session can resume without rediscovering the project.

36. v1.2 Final Release Chain

Market Radar -> Opportunity Score -> Creative Brief -> Originality/IP Gate -> CIR -> Compiler -> Digital Twin -> Reverse Compiler -> Adversarial QA -> Physical Test if required -> PDF/Charts/Video Assets -> Asset Truth -> Pricing -> Listing/SEO -> Policy Gate -> Quality Release Certificate -> Shadow/Staging Check -> Store Operator -> Marketing -> Customer Experience -> Review/Incident Learning -> Portfolio Reallocation.

Living-plan rule: future material improvements become v1.3, v1.4, v2.0, etc. Preserve the prior version and changelog. Do not scatter canonical strategy across chat messages.